

1 Introduction

The spread of epidemics in networks, such as diseases or computer viruses, presents critical challenges with significant societal impact [16, 19]. Effective immunization strategies aim to vaccinate a select subset of nodes to minimize this spread [20, 5], especially when resources are initially scarce, as seen during the early stages of pandemics like COVID-19.

In this work, we tackle the problem of *adaptive vaccination*, focusing on selecting the optimal nodes for immunization based on the current state of infection spread, during times of scarce vaccine availability. We hypothesize that **PageRank**’s centrality, a well-established measure for identifying influential nodes, can effectively guide vaccination efforts. PageRank has traditionally been applied to web ranking [17], but we extend its use to vaccination strategy optimization.

We evaluate the performance of Random, Degree-based, PageRank-based, and NetShield algorithms [24] under both the Independent Cascade (IC) and Susceptible-Infected-Recovered (SIR) models [9, 10]. Our results on real-world networks demonstrate that the PageRank-based strategy consistently outperforms other heuristics in minimizing the spread of epidemics.

2 Problem Formulation

The goal of this work is to design a vaccination strategy that adapts to both the network’s structure and the current state of an ongoing epidemic. Given a network $G = (V, E)$, where V represents the set of nodes and E represents the set of edges, let $\mathcal{S} \subseteq V$ denote the set of infected (‘sick’) nodes at a particular point in time. With a limited vaccination budget of k , our objective is to identify a set of uninfected (‘healthy’) nodes, denoted by $\mathcal{H}$, to vaccinate in order to minimize future infection spread.

We define the adaptive vaccination problem as finding a set $\mathcal{H} \subseteq V \setminus \mathcal{S}$ of size $|\mathcal{H}| = k$ such that the expected number of infected nodes at the end of the epidemic, denoted as $\mathbb{E}_G^{S, \mathcal{H}}$, is minimized. This can be formalized as:

$$\mathcal{H}^* = \arg \min_{\mathcal{H} \subseteq V \setminus \mathcal{S}, |\mathcal{H}| = k} \mathbb{E}_G^{S, \mathcal{H}}.$$

Over large networks imitating realistic scenarios, we hypothesized that an exact solution to this problem would be computationally challenging. Therefore, we explore heuristic strategies that leverage network centrality measures to guide the vaccination process.

3 Background

Given that our problem formulation from Section 2 allows us to reason about the spread of diseases in human populations and computer viruses in digital networks, we provide a brief overview of the relevant concepts and models.

3.1 Epidemic Models

We consider two epidemic models, each suited to different disease dynamics:

Susceptible-Infected-Recovered (SIR) Model [10] The SIR model is commonly used to study disease spread in networks [2, 7, 23]. It models infectious diseases with varying recovery rates, and nodes transition through three states: Susceptible, Infected, and Recovered. Susceptible nodes can become infected, infected nodes can infect susceptible neighbors, and infected nodes recover with probability δ , becoming immune.

Independent Cascade (IC) Model [9] A variant of the SIR model, the IC model is used to simulate information diffusion in networks [9]—by allowing nodes to infect neighbors only once.

3.2 Vaccination Strategies

Vaccination strategies in networks have been studied [5, 8]. Preemptive methods use centrality measures, like degree or betweenness, to select vaccination nodes [11]. In contrast, adaptive approaches consider the epidemic’s current state for decision-making [18].

The NetShield algorithm [24] selects nodes to minimize the spectral radius of the adjacency matrix, reducing vulnerability to epidemic spread. Our work differs by exploring PageRank centrality as a heuristic for adaptive vaccination.

4 PageRank Centrality

PageRank is a measure of node importance based on the idea that connections from important nodes contribute more to the importance of a node [17]. The PageRank score $PR(i)$ of node i is defined recursively as:

$$PR(v) = \frac{\epsilon}{N} + (1 - \epsilon) \sum_{(v', v) \in E} \frac{PR(v')}{\text{out-degree}(v')}$$

where ϵ is the damping factor (typically $1/7$) and N is the total number of nodes. We leverage PageRank to identify well-connected, central nodes for vaccination, as these are likely to influence epidemic spread.

5 Methodology

5.1 Epidemic Simulation

We use the SIR and IC models detailed in Section 3.1 to simulate the spread of epidemics on real-world networks under different disease dynamics. In particular, we use the following propagation models:

Model	Infection Probability (p)	Recovery Probability (δ)	Modeling Disease
SIR	0.8	0.1	Ebola-like (due to uncontrollable infection)
IC	0.4	N/A	COVID-19-like (due to near-instant quarantining)

5.2 Vaccination Strategies

We implement and compare the following strategies:

Strategy	Description	Function in Code
No Vaccination	No nodes are vaccinated.	N/A
Random	Randomly select k healthy nodes to vaccinate.	<code>random_strategy</code>
NetShield	Use the NetShield algorithm [24] to select nodes that minimize the spectral radius.	<code>netshield</code>
Degree-based	Select the k healthy nodes with the highest degrees. A degenerate case of PageRank, as discussed in class.	<code>degree_strategy</code>
PageRank-based	Select the k healthy nodes with the highest PageRank scores.	<code>pagerank_strategy</code>

5.3 Data Modeling

To apply the vaccination strategies to real-world network data, we first modeled the datasets to fit these strategies. We utilized and modified code from previous assignments related to network representation and centrality measures.

All networks are undirected graphs, where each line corresponds to an edge between two node IDs. Each edge is represented as two directed edges to capture the undirected nature.

Dealing With Sinks Since PageRank’s convergence relies on preserving the L1-norm of 1 for the ranks vector, we address nodes with no outgoing edges that would leak rank. We assume each sink node has an edge to itself during PageRank computation, thus preserving the graph’s structure while adjusting the computation.

5.4 Experimental Setup

Social Networks We conduct experiments on four real-world social and peer-to-peer networks, detailed in [Appendix B](#).

Epidemic Propagation Models Configurations We simulate both the IC and SIR models on each network to model different disease dynamics, as discussed in [Section 3.1](#). The configuration parameter ranges are varied based on the network size. We vary the vaccination budget size to evaluate the strategies’ performance under different resource constraints.

Network	Initial Infection Size ($ S $)	Vaccination Budget (k)	Simulations/Configuration
Facebook, LastFM, Gnutella	100	10,30,50,70,90,110,130,150	200
Brightkite	400	10,50,100,150,200,250,300,350	50

Evaluation Metric We evaluate each strategy based on the **average number of healthy nodes** remaining after the epidemic. A higher count of healthy nodes indicates a more effective vaccination strategy, averaged across simulations per configuration.

Code The `run_experiment` in `project.py` encompasses the entire experiment process, including epidemic simulation, vaccination, and result evaluation. It runs experiments for each network and vaccination strategy, with the `main` function already set to call it.

6 Experimental Results, Discussion, and Conclusions

Detailed figures for each network under both epidemic models are provided in the [Artifacts Appendix](#) (see Figures 1 and 2).

Our experiments demonstrate that the PageRank-based vaccination strategy significantly reduces epidemic spread compared to other heuristics across various networks and models. Additionally, we observe a notable runtime improvement over the NetShield algorithm, which requires solving an optimization problem to minimize the spectral radius.

Discussion The effectiveness of the PageRank-based strategy may stem from its ability to identify highly connected and central nodes, which are more likely to influence a large portion of the network [17]. Furthermore, PageRank’s imitation of a random walk closely relates to disease spread in networks. Our results indicate that PageRank centrality provides a practical and computationally efficient heuristic.

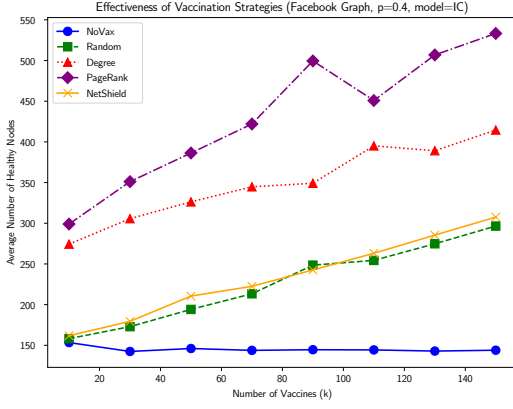
These findings have significant implications for managing epidemics, particularly when vaccine resources are limited. By vaccinating central nodes, health authorities can effectively curb the spread of infectious diseases. Strategies like PageRank-based are crucial for prioritizing vaccination targets during early outbreak stages and applies to both human social networks and digital networks, enhancing public health interventions and informing policy decisions.

Conclusions We show that PageRank centrality is an effective heuristic for node vaccination in the adaptive vaccination problem. Our experiments on real-world networks validate our hypothesis, demonstrating the potential of using PageRank for adaptive vaccination in network epidemics. Future work may explore more complex dynamic vaccination strategies and investigate combining PageRank with other centrality measures to enhance vaccination effectiveness.

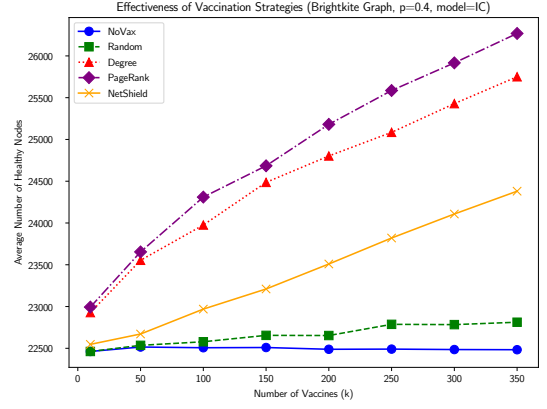
A Artifacts

In this appendix, we present the experimental results in the form of two composite figures, each containing four subfigures corresponding to the different networks analyzed under each epidemic model.

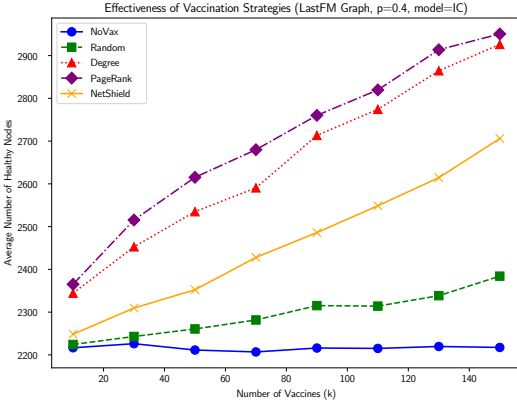
A.1 Independent Cascade Model Results (COVID-19-like)



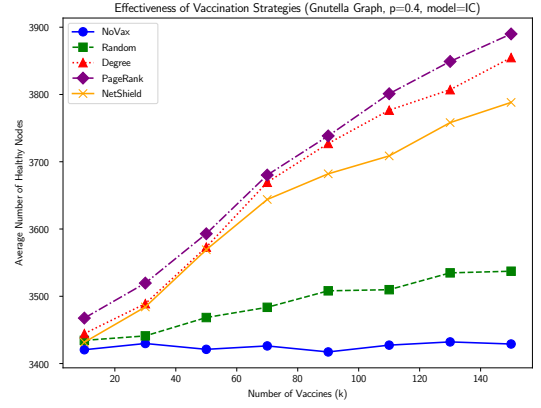
(a) Facebook Network



(b) Brightkite Network



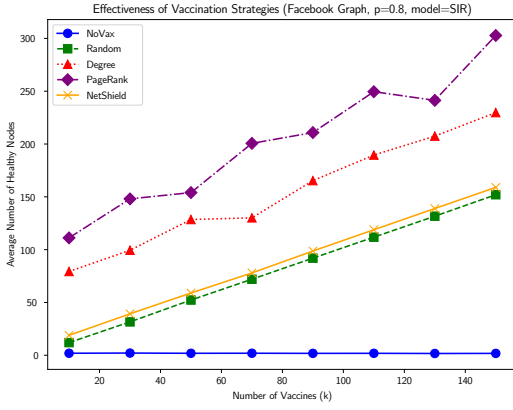
(c) LastFM Network



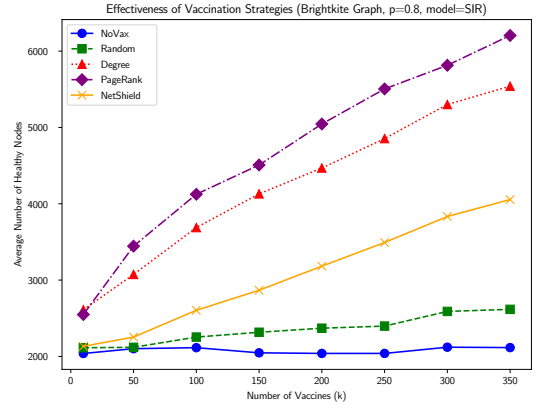
(d) Gnutella Network

Figure 1: Effectiveness of Vaccination Strategies under the Independent Cascade Model ($p=0.4$)

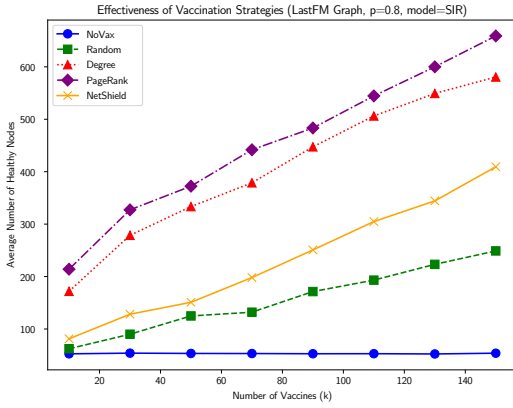
A.2 SIR Model Results (Ebola-like)



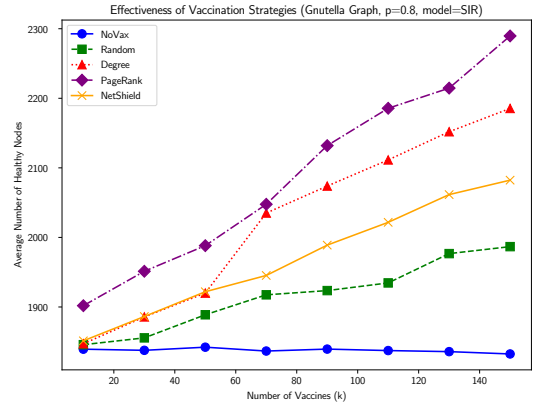
(a) Facebook Network



(b) Brightkite Network



(c) LastFM Network



(d) Gnutella Network

Figure 2: Effectiveness of Vaccination Strategies under the SIR Model ($p=0.8$, $\delta=0.1$)

B Datasets

We used the following real-world network datasets in our experiments, taken from the [14, Stanford Network Analysis Project (SNAP)]:

Dataset	Description	Nodes	Edges
Facebook Social Network [6]	A network representing friendships on Facebook [15].	4,039	88,234
Brightkite Friendship Network [3]	A network from the Brightkite location-based social networking platform. We used the friendship network for our experiments [4].	58,228	214,078
LastFM Music Social Network [12]	A network representing friendships on LastFM, a music recommendation platform [22].	7,624	27,806
Gnutella Peer-to-Peer Network [1]	A network of Gnutella peer-to-peer file sharing connections [13, 21]. It serves as a model for online virus spread, expanding the scope of our experiments.	6,301	20,777

References

- [1] Gnutella 08. Gnutella 08 Data. <https://snap.stanford.edu/data/p2p-Gnutella08.html>, August 2002.
- [2] Roy M Anderson and Robert M May. *Infectious diseases of humans: dynamics and control*. Oxford university press, 1991.
- [3] Brightkite. Brightkite Data. <https://snap.stanford.edu/data/loc-Brightkite.html>, October 2010.
- [4] Eunjoon Cho, Seth A Myers, and Jure Leskovec. Friendship and mobility: user movement in location-based social networks. In *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 1082–1090, 2011.
- [5] Reuven Cohen, Shlomo Havlin, and Daniel Ben-Avraham. Efficient immunization strategies for computer networks and populations. *Physical review letters*, 91(24):247901, 2003.
- [6] Facebook. Facebook Data. <https://snap.stanford.edu/data/ego-Facebook.html>, June 2012.
- [7] Herbert W Hethcote. The mathematics of infectious diseases. *SIAM review*, 42(4):599–653, 2000.
- [8] Petter Holme, Beom Jun Kim, Chang No Yoon, and Seung Kee Han. Attack vulnerability of complex networks. *Physical review E*, 65(5):056109, 2002.

- [9] David Kempe, Jon Kleinberg, and Éva Tardos. Maximizing the spread of influence through a social network. In *Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 137–146, 2003.
- [10] William Ogilvy Kermack and Anderson G McKendrick. A contribution to the mathematical theory of epidemics. *Proceedings of the royal society of london. Series A, Containing papers of a mathematical and physical character*, 115(772):700–721, 1927.
- [11] Maksim Kitsak, Lazaros K Gallos, Shlomo Havlin, Fredrik Liljeros, Lev Muchnik, H Eugene Stanley, and Hernán A Makse. Identification of influential spreaders in complex networks. *Nature physics*, 6(11):888–893, 2010.
- [12] LastFM. LastFM Data. <https://snap.stanford.edu/data/feather-lastfm-social.html>, March 2020.
- [13] Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. Graph evolution: Densification and shrinking diameters. *ACM transactions on Knowledge Discovery from Data (TKDD)*, 1(1):2–es, 2007.
- [14] Jure Leskovec and Andrej Krevl. SNAP Datasets: Stanford large network dataset collection. <http://snap.stanford.edu/data>, June 2014.
- [15] Jure Leskovec and Julian McAuley. Learning to discover social circles in ego networks. *Advances in neural information processing systems*, 25, 2012.
- [16] Mark EJ Newman. The structure and function of complex networks. *SIAM review*, 45(2):167–256, 2003.
- [17] Lawrence Page. The pagerank citation ranking: Bringing order to the web. Technical report, Technical Report, 1999.
- [18] Romualdo Pastor-Satorras, Claudio Castellano, Piet Van Mieghem, and Alessandro Vespignani. Epidemic processes in complex networks. *Reviews of modern physics*, 87(3):925–979, 2015.
- [19] Romualdo Pastor-Satorras and Alessandro Vespignani. Epidemic spreading in scale-free networks. *Physical review letters*, 86(14):3200, 2001.
- [20] Romualdo Pastor-Satorras and Alessandro Vespignani. Immunization of complex networks. *Physical review E*, 65(3):036104, 2002.
- [21] Matei Ripeanu, Ian Foster, and Adriana Iamnitchi. Mapping the gnutella network: Properties of large-scale peer-to-peer systems and implications for system design. *arXiv preprint cs/0209028*, 2002.
- [22] Benedek Rozemberczki and Rik Sarkar. Characteristic Functions on Graphs: Birds of a Feather, from Statistical Descriptors to Parametric Models. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM '20)*, page 1325–1334. ACM, 2020.
- [23] Devavrat Shah and Tauhid Zaman. Rumors in a network: Who’s the culprit? *IEEE Transactions on information theory*, 57(8):5163–5181, 2011.
- [24] Hanghang Tong, B Aditya Prakash, Charalampos Tsourakakis, Tina Eliassi-Rad, Christos Faloutsos, and Duen Horng Chau. On the vulnerability of large graphs. In *2010 IEEE international conference on data mining*, pages 1091–1096. IEEE, 2010.